

Hyperspherical Fitness Functions for Many-Objective Optimization*

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Abstract

Many-objective optimization problems (MaOPs) with more than three objectives pose fundamental challenges for evolutionary algorithms based on Pareto dominance. As dimensionality increases, selection pressure collapses because nearly all solutions become mutually non-dominated. We introduce *hyperspherical fitness functions*, a family of angular scalarization methods that project objective vectors onto the unit hypersphere and measure fitness as angular distance from a target direction. This approach provides a Pareto-independent selection signal, reducing complexity from $O(MN^2)$ to $O(MN)$ while maintaining selection pressure at extreme objective counts. We present two variants—HFF-Balanced and HFF-TrueNorth—with complementary geometric properties. Experimental evaluation on DTLZ1–7 and WFG1–9 benchmarks from 10 to 1000 objectives reveals that HFF excels at identifying exceptional individual solutions (outperforming NSGA-II on 14 of 16 problem types, and 92% of problem-objective pairs, on CDF-corrected angular distance) while NSGA-II retains superiority on Pareto front coverage. We introduce *Arctic Circles*—angular level sets on the hypersphere—as a geometric framework for understanding solution equivalence under angular fitness.

CCS Concepts

• Computing methodologies → Genetic algorithms; Optimization algorithms.

Keywords

many-objective optimization, evolutionary algorithms, hyperspherical geometry, fitness functions, NSGA-II, scalability

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1 Introduction

Many-objective optimization problems (MaOPs)—those involving four or more conflicting objectives [8]—arise naturally in engineering design, portfolio optimization, neural architecture search, and numerous other domains. Traditional multi-objective evolutionary algorithms (MOEAs) based on Pareto dominance, such as NSGA-II [11], suffer from a well-documented failure mode as objective count increases: selection pressure collapses because nearly all solutions become mutually non-dominated [17, 18].

The core issue is statistical: in high-dimensional objective spaces, the probability that one random solution dominates another approaches zero [13]. With m objectives, a solution must be no worse on all m dimensions and strictly better on at least one. As m grows, this becomes exponentially unlikely for typical solution distributions. The result is that Pareto-based selection degenerates to random selection, and evolutionary progress stalls. Recent surveys identify algorithmic scalability as a key open challenge, noting that “it is not trivial to produce a selection mechanism that is not Pareto-based, decomposition-based or indicator-based” [7].

In this paper, we introduce *hyperspherical fitness functions*, a family of angular scalarization methods that delivers such a selection mechanism. We propose abandoning Pareto dominance, instead projecting objective vectors onto the unit hypersphere and measuring fitness as an angular distance between the objective and target points. This distance-oriented measurement maintains meaningful selection pressure even at extreme objective counts, enabling HFF based methods to find solutions to previously intractable problems.

This work distinguishes two optimization goals: identifying exceptional individual solutions under concentration of measure, and approximating the Pareto front as a set. Hyperspherical fitness targets the former; dominance-based algorithms target the latter.

Contributions. This paper makes the following contributions:

- We introduce hyperspherical fitness functions based on angular scalarization, providing a mathematically principled alternative to Pareto dominance for many-objective optimization.
- We define two complementary variants—HFF-Balanced and HFF-TrueNorth—suited to different problem geometries.
- We demonstrate problem-geometry-dependent performance advantages over NSGA-II across the DTLZ and WFG benchmark suites.
- We show scalability to 1000 objectives with CDF-corrected analysis accounting for concentration of measure.
- We introduce Hyperspherical IGD (HIGD), a CDF-corrected metric for evaluating convergence on WFG4–9 (unit hypersphere Pareto fronts).

- We provide analytical Pareto front sampling strategies for WFG1–3 that enable Euclidean IGD computation at arbitrary objective counts, bypassing the exponential scaling of standard enumeration methods.
- We introduce the concept of *Arctic Circles*—angular level sets that reveal the geometric structure of Pareto-equivalent trade-offs.

2 Related Work

2.1 Many-Objective Optimization

Three main strategies have emerged to address selection pressure collapse. *Reference-based methods* such as NSGA-III [10] and RVEA [6] use predefined reference directions to maintain diversity and guide convergence. *Decomposition approaches* like MOEA/D [23] transform the multi-objective problem into multiple scalar subproblems. *Indicator-based methods* including IBEA [26], SMS-EMOA [3], and HypE [1] use quality indicators such as hypervolume to rank solutions.

These methods help, but they remain bound to Pareto dominance—and recent theory shows why that’s a problem. Weak Pareto boundaries induce dominance-resistant solutions, with resistance growing as $\Theta(\Delta^v)$. As a result, elimination becomes nearly impossible [24].

Perhaps the problem is Pareto dominance itself. By removing it from selection entirely, we can focus on fitness measures that remain discriminative as objectives grow—even to 1000.

2.2 Scalarization and Indicator-Based Methods

Weighted sum methods suffer from well-known limitations on non-convex fronts [9]. Indicator-based MOEAs use quality indicators for selection: IBEA [26] introduced this paradigm, with the hypervolume [25] being the only Pareto-compliant unary indicator [27]. However, hypervolume computation grows exponentially with objectives [2], motivating approximations like HypE [1] and alternative indicators such as R2 [5].

Our hyperspherical fitness can be viewed as a geometric scalarization that naturally handles arbitrary objective counts without requiring weight vectors, reference points, or expensive indicator calculations.

2.3 Industrial Motivation: Digital Twin Systems

In industrial digital twin and cyber-physical systems, optimization is embedded within a continuous closed-loop decision process rather than treated as an offline task. Such systems integrate sensing, state estimation, decision-making, and actuation in real time, forming a repeated observe–orient–decide–act cycle [14, 20, 22]. In these settings, actionable recommendations and rapid convergence are more valuable than large sets of mutually non-dominated solutions, motivating convergence-oriented many-objective optimization.

3 Hyperspherical Fitness Functions

We now present the mathematical foundation of hyperspherical fitness functions.

3.1 Hyperspherical Projection

Given an objective vector $\mathbf{x} = (x_1, \dots, x_m) \in \mathbb{R}^m$, where m is the number of objectives and all are to be minimized, we project onto the unit hypersphere:

$$\mathbf{y} = \frac{\mathbf{x}}{\|\mathbf{x}\|} \quad (1)$$

The projected vector $\mathbf{y} = (y_1, \dots, y_m)$ lies on the unit $(m-1)$ -sphere, satisfying $\|\mathbf{y}\| = 1$. Since we divide by a positive scalar, signs are preserved, allowing the method to handle objectives that may take negative values; for standard minimization benchmarks where $x_i \geq 0$, the projection lies in the positive orthant.

Energy interpretation. Define the total energy $E = \|\mathbf{x}\|^2 = \sum_{i=1}^m x_i^2$. Then $y_i = x_i/\sqrt{E}$, and each squared coordinate $y_i^2 = x_i^2/E$ represents the fractional energy contributed by objective i . Solutions with balanced trade-offs have coordinates near $(1/\sqrt{m}, \dots, 1/\sqrt{m})$, while solutions dominated by a single objective concentrate near the coordinate axes.

The energy E captures total solution magnitude: high-energy solutions are far from the origin, while low-energy solutions approach ideal minimization. This distinction motivates our two fitness variants—one rewards balanced energy distribution, the other rewards minimal total energy.

3.2 Fitness Variants

We define fitness as the angular distance from a target direction on the hypersphere.

3.2.1 HFF-Balanced (HFF-B). The Balanced variant targets equal energy distribution across objectives:

$$\mathbf{p}_B = \frac{1}{\sqrt{m}}(1, 1, \dots, 1) \in \mathbb{R}^m \quad (2)$$

Fitness is the angular distance from this balanced direction:

$$\theta_B = \arccos(\mathbf{y} \cdot \mathbf{p}_B) = \arccos\left(\frac{1}{\sqrt{m}} \sum_{i=1}^m y_i\right) \quad (3)$$

Geometric interpretation. HFF-B rewards solutions where each objective contributes equally to total energy. It is well-suited to problems with symmetric or convex Pareto fronts where balanced trade-offs are desirable.

3.2.2 HFF-TrueNorth (HFF-TN). HFF-B is agnostic to total energy. HFF-TrueNorth directly rewards minimization toward zero energy.

Projective embedding. The projection $\mathbf{y} = \mathbf{x}/\|\mathbf{x}\|$ is undefined at $\mathbf{x} = \mathbf{0}$ —the ideal solution. The classical remedy from projective geometry is to append a constant before normalization, converting Euclidean distance into angular distance while remaining well-defined at the origin [21].

We adopt a scaled variant. Given $\mathbf{x} \in \mathbb{R}^m$ with energy $E = \|\mathbf{x}\|^2$, define:

$$e = \max\left(0, 1 - \frac{E}{m}\right) \in [0, 1] \quad (4)$$

The augmented projection $\mathbf{y}' = (y_1\sqrt{1-e^2}, \dots, y_m\sqrt{1-e^2}, e) \in S^m$ places zero-energy solutions at the north pole $\mathbf{p}_{TN} = (0, \dots, 0, 1)$. Fitness is:

$$\theta_{TN} = \arccos(e) \quad (5)$$

When $E = 0$, the scaling factor $\sqrt{1 - e^2}$ vanishes, yielding $\mathbf{y}' = \mathbf{p}_{TN}$ and $\theta_{TN} = 0$. When $E \geq m$, we have $e = 0$ and $\theta_{TN} = \pi/2$ —the equator represents high-energy solutions.

3.3 Selection Mechanism

Hyperspherical fitness *replaces* Pareto dominance ranking entirely. Selection proceeds via standard tournament selection on the scalar angular fitness θ , where lower values are preferred (closer to the north pole).

Arctic Circles. Solutions with identical angular distance θ lie on a level set we term an *Arctic Circle*—a hyperspherical cap centered on the north pole. All solutions on the same Arctic Circle are considered equivalent under hyperspherical fitness, analogous to Pareto indifference but geometrically structured.

Importantly, this equivalence is *equitable* across objectives. Because angular distance depends only on the normalized contribution of each objective, no single objective—or group of objectives—is privileged. All objectives contribute symmetrically to fitness, and alternative trade-offs that preserve this balance are treated as equally valid. In multi-agent interpretations where objectives represent agent preferences, Arctic Circles therefore encode a notion of fairness: solutions on the same level set favor no particular agent.

Tournament selection among Arctic Circle members acts as stochastic tie-breaking, maintaining diversity along the front without introducing objective bias.

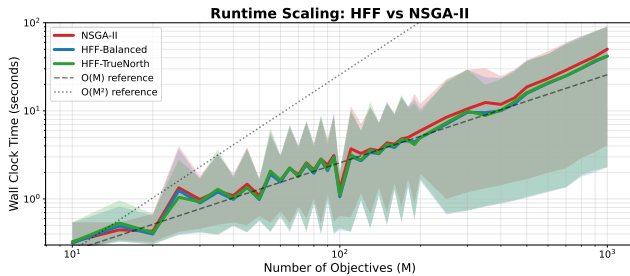


Figure 1: Runtime scaling with objective count (log-log scale). All algorithms follow $O(M)$ scaling; the theoretical $O(M^2)$ reference line shows where quadratic growth would appear. HFF achieves superior convergence without computational overhead.

4 Computational Complexity

NSGA-II. Non-dominated sorting has worst-case complexity $O(MN^2)$ for M objectives and N population members [16]. While efficient implementations reduce average-case performance, the quadratic dependence on N remains.

Hyperspherical fitness. Computing angular fitness for all N solutions requires:

- Energy calculation: $O(M)$ per solution, $O(MN)$ total
- Projection and dot product: $O(M)$ per solution, $O(MN)$ total
- Arccos: $O(1)$ per solution, $O(N)$ total

Total complexity: $O(MN)$ —linear in both population size and objective count.

Figure 1 confirms this analysis empirically. We compare against pymoo’s optimized, vectorized NSGA-II implementation; all three algorithms exhibit similar $O(M)$ scaling with objective count, demonstrating that HFF achieves superior convergence without computational overhead.

5 Experimental Setup

5.1 Benchmarks

We evaluate on two standard many-objective benchmark suites:

- **DTLZ suite:** DTLZ1–7 [12], covering problems with linear, concave, convex, and disconnected Pareto fronts.
- **WFG suite:** WFG1–9 [15], featuring separable and non-separable problems with various front geometries.

We test across objective counts from 10 to 1000, with denser sampling in the 10–200 range where algorithm behavior diverges most significantly.

5.2 Algorithms Compared

Table 1: Algorithms evaluated in the experimental study.

Algorithm	Description
NSGA-II	Pareto dominance + crowding distance
HFF-Balanced	Hyperspherical fitness, $\mathbf{p} = (1/\sqrt{m}, \dots, 1/\sqrt{m})$
HFF-TrueNorth	Hyperspherical fitness, augmented space

Table 1 summarizes the algorithms. All three use identical genetic operators, differing only in the selection mechanism.

5.3 Parameters

Table 2: Experimental parameters.

Parameter	Value
Population size	100
Generations	100
Independent runs	≥ 31 per configuration
Selection	Binary tournament
Crossover	SBX ($\eta_c = 15$)
Mutation	Polynomial ($\eta_m = 20$)
Objectives	10–1000

Table 2 lists experimental parameters. All experiments were implemented using pymoo [4].

5.4 Evaluation Metrics

CDF-corrected angular distance is the primary metric for evaluating hyperspherical fitness, as it measures individual solution extremeness under concentration of measure. HIGD and Euclidean IGD are diagnostic metrics for Pareto front coverage.

Raw angular distance. During evolution, fitness is the angular distance $\theta \in [0, \pi]$ from the north pole to the best solution found.

CDF-corrected angular distance. Raw angular distances are not comparable across objective counts due to *concentration of measure*—random points on high-dimensional hyperspheres concentrate near the equator. To enable fair comparison, we apply a CDF correction based on the known distribution of angular distances for uniform random points.

For a uniform random point on the $(m - 1)$ -sphere, the angular distance θ from any fixed pole follows:

$$P(\theta \leq t) = I_{\sin^2(t)}\left(\frac{m-1}{2}, \frac{1}{2}\right) \quad (6)$$

where $I_x(a, b)$ is the regularized incomplete beta function.

The corrected value represents a percentile rank: values near 0 indicate solutions in regions essentially inaccessible to random search, while values near 0.5 indicate equator-like (random) performance.

Hyperspherical IGD (HIGD). For problems with unit hypersphere Pareto fronts (WFG4–9) [15], we introduce HIGD: the mean CDF-corrected angular distance from each reference point on the Pareto front to its nearest solution. Reference points are sampled uniformly on the positive orthant of the unit hypersphere.

Euclidean IGD. For problems with non-spherical Pareto fronts (WFG1–3), we use standard Euclidean IGD with analytically sampled reference fronts, bypassing the exponential enumeration that limits standard implementations to approximately 15 objectives.

6 Results

6.1 Concentration of Measure Baseline

Figure 2 shows raw angular distance (radians) without CDF correction, comparing NSGA-II against HFF variants which appear to consistently outperform apart from the DTLZ7 challenge, which confounds HFF methods.

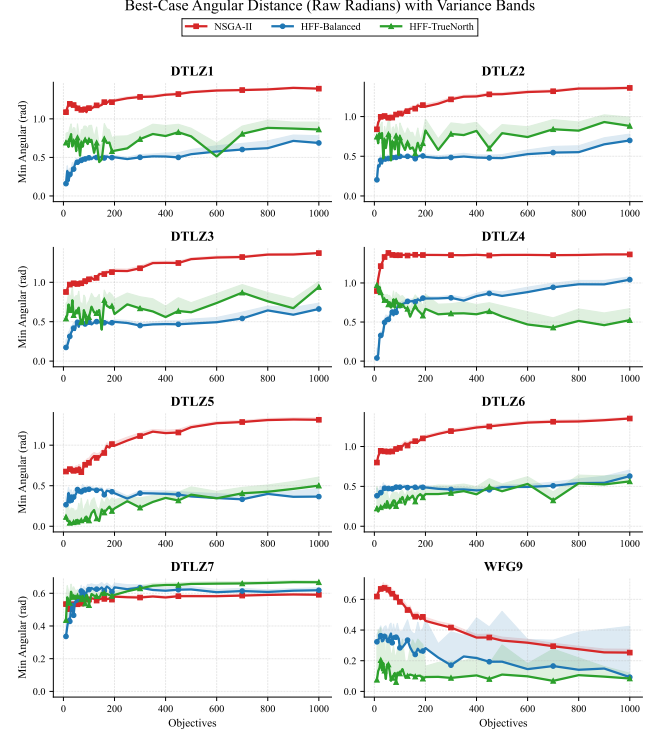


Figure 2: Raw angular distance (radians) for representative problems. All algorithms trend toward $\pi/2$ due to concentration of measure, necessitating CDF correction.

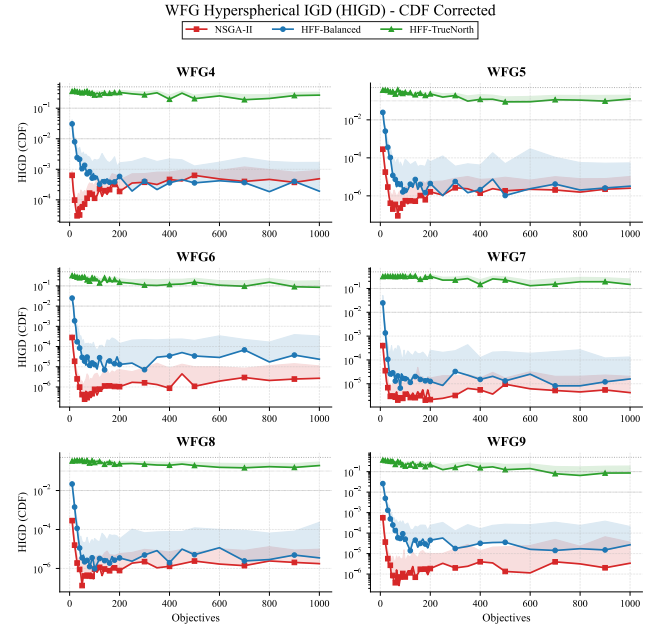


Figure 4: Hyperspherical IGD (CDF-corrected) for WFG4–9. Lower values indicate better coverage of the Pareto front. We find NSGA-II clearly outperforms HFF-TN as measured by HIGD, but note HFF-B tracks NSGA-II which is a more positive finding, and unexpected.

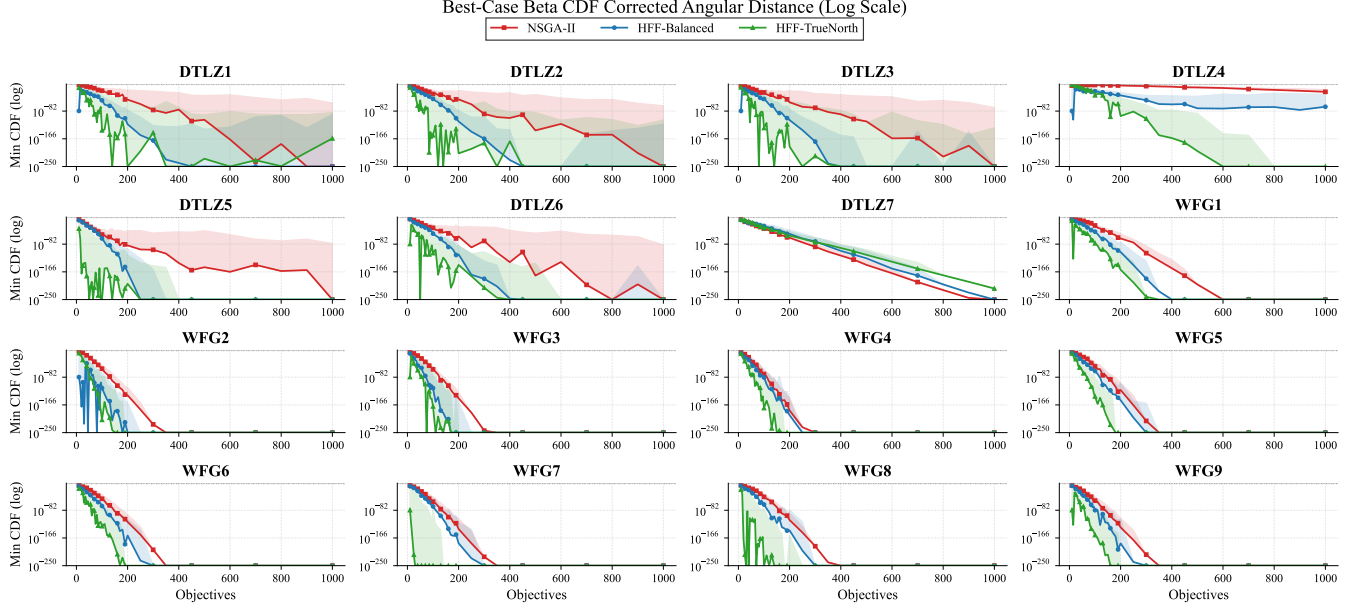


Figure 3: CDF-corrected angular distance (log scale). Lower is better. HFF-TrueNorth exerts magnitudes more selection pressure than NSGA-II across all problems except DTLZ7 (disconnected front), though with high variance. HFF-Balanced also consistently outperforms NSGA-II with lower variance, but lags behind HFF-TN in solution quality.

6.2 CDF-Corrected Scaling

Figure 3 presents CDF-corrected angular distance across objective counts. This is the key result of our study. The charts show that across the benchmarks HFF methods measurably surface high-quality solutions when evaluated against the minimization target. HFF-TN outcompetes the other algorithms in 14 of 16 benchmarks at dimensions between 10 and approximately 300.

A key observation is that performance depends on problem geometry. These results offer insight into which problem structures suit HFF methods best. Conversely, for an unknown problem, HFF performance may reveal underlying geometric characteristics.

We note that DTLZ4 (biased density toward axes) drives NSGA-II to produce results barely better than random, even after CDF correction. DTLZ4 also poses difficulties for HFF-B, but not for HFF-TN—suggesting TrueNorth’s minimization-directed fitness cuts a path through the biased densities.

We point out the axis of the chart is logarithmic and has an artificial floor introduced by us at 10^{-250} to allow our charting to render. This scale indicates that for most problems HFF-TN is able to deliver a solution which is magnitudes closer in corrected terms to the minimization target than those found by HFF-B or NSGA-II. This finding is fairly consistent when the number of objectives is between 10 and 200. Interestingly, the corrected CDF data illustrate that at over 400 objectives NSGA-II is doing better than the unadjusted data suggests.

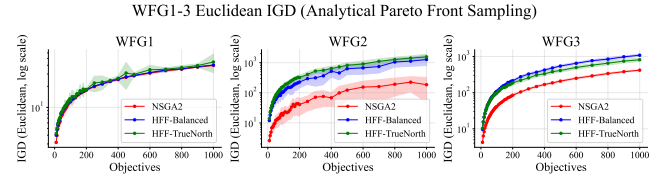


Figure 5: Euclidean IGD for WFG1–3 using analytical Pareto front sampling. NSGA-II also clearly outperforms HFF methods, in line with the findings in Figure 4. WFG2, having a disconnected convex front, highlights this unambiguously.

6.3 HIGD and IGD Results

Figures 4 and 5 present IGD results. As expected, hyperspherical fitness performs poorly on these Pareto coverage metrics, confirming that HFF optimizes individual-solution extremeness rather than set-level coverage.

6.4 Statistical Significance

Figure 6 presents a performance comparison heatmap comparing both HFF variants against NSGA-II across all problem-objective combinations. Statistical significance is assessed using the Mann-Whitney U test; cells marked with dots indicate $p < 0.05$.

HFF-Balanced significantly outperforms NSGA-II on the majority of problem-objective combinations, with the advantage most pronounced at high objective counts. The primary exceptions are:

- DTLZ5: Degenerate front geometry.
- DTLZ6: Difficult convergence landscape.

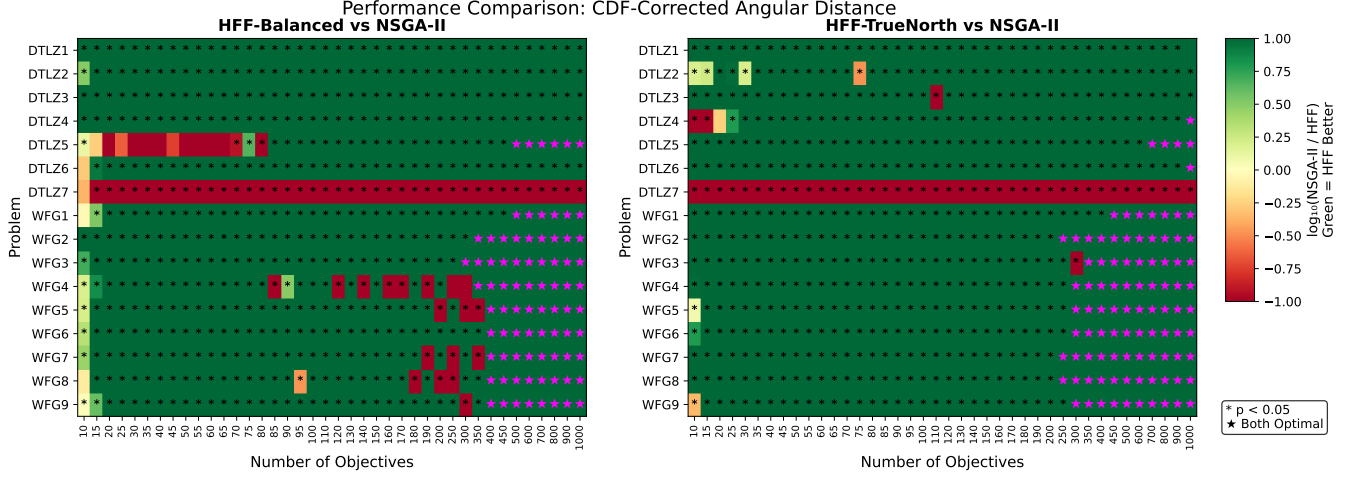


Figure 6: Performance comparison using CDF-corrected angular distance. Green indicates HFF outperforms NSGA-II; red indicates NSGA-II wins. Dots mark statistical significance ($p < 0.05$). HFF-Balanced dominates across most benchmarks at all objective counts. Notable exceptions: DTLZ5 (degenerate front) and DTLZ6 (difficult convergence landscape).

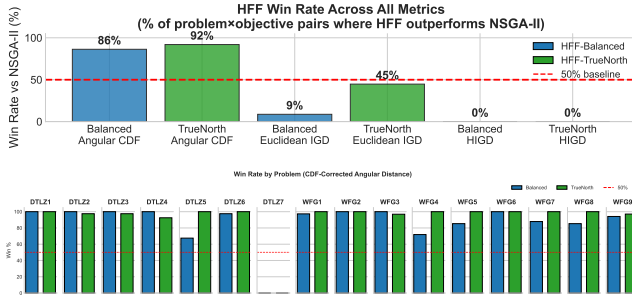


Figure 7: Top: HFF win rate across all metrics. On CDF-corrected angular distance, HFF-Balanced wins 86% and HFF-TrueNorth wins 92% of problem-objective pairs. Bottom: Win rate breakdown by problem shows consistent dominance across DTLZ1–4 and WFG1–9, with reduced performance on DTLZ5–7 (degenerate/disconnected fronts).

6.5 Aggregate Performance

Figure 7 summarizes win rates across all problem-objective configurations. On CDF-corrected angular distance, both HFF variants decisively outperform NSGA-II: HFF-Balanced wins 86% and HFF-TrueNorth wins 92% of all problem-objective pairs. In contrast, Euclidean IGD shows near-parity—illustrating why traditional metrics fail to capture performance differences at high dimensions.

6.6 HIGD Analysis for WFG4–9

For WFG4–9, which have unit hypersphere Pareto fronts, we compute Hyperspherical IGD (HIGD). HIGD replaces the Euclidean distance in standard IGD with CDF-corrected angular distance: for each solution, we compute the angular distance to the nearest reference point on the Pareto front, then apply the Beta CDF correction (Equation 6) to account for concentration of measure. Reference

points are sampled uniformly on the unit hypersphere using the Muller-Marsaglia method [19] (Algorithm 3). Figure 4 shows HIGD results. As expected, hyperspherical fitness performs poorly on HIGD, confirming that it does not approximate Pareto fronts. This result validates that HFF optimizes individual-solution extremeness rather than set-level coverage.

6.7 Euclidean IGD for WFG1–3

WFG1–3 have non-spherical Pareto fronts (scaled simplex for WFG1–2, degenerate line for WFG3). Standard implementations cannot generate reference fronts beyond approximately 15 objectives due to exponential enumeration. We bypass this limitation through analytical sampling based on the known front geometries (Algorithms 1–2). Figure 5 shows Euclidean IGD using these analytically sampled reference fronts.

7 Discussion

Complementary, not replacement. HFF does not replace NSGA-II or other Pareto-based algorithms. Our IGD results (Figures 4–5) confirm that NSGA-II remains superior for Pareto front coverage. Rather, HFF offers a complementary lens: the ability to rank solutions—including those that are Pareto-equivalent—based on geometric criteria. This is valuable when Pareto dominance produces large incomparable sets and decision-makers need actionable rankings.

Different objectives. CDF-corrected angular distance and HIGD measure fundamentally different objectives: the former evaluates extremeness and decision relevance of individual solutions under concentration of measure, while the latter evaluates set-level Pareto front coverage. HFF is designed for the former and intentionally sacrifices the latter.

What equivalence means. For HFF-Balanced, solutions on the same Arctic Circle share identical angular distance from the balanced pole. Crucially, this equivalence is based on *directional balance*, not magnitude—two solutions can be deemed equivalent even if one is far from minimizing the objectives. The metric rewards balanced trade-offs across objectives, not proximity to zero. When goals are clear and unambiguous, this property is useful for exploration, but eventually a magnitude-aware selection (or HFF-TrueNorth) may be needed.

Fairness interpretation. Arctic Circles encode a geometric notion of fairness: all objectives contribute symmetrically to fitness, and no single objective is privileged. In multi-agent settings where objectives represent stakeholder preferences, solutions on the same Arctic Circle treat all stakeholders equitably. This fairness property may be valuable in resource allocation, multi-user systems, or any domain where equitable treatment of competing goals matters more than raw minimization.

HFF-TrueNorth vs HFF-Balanced. TrueNorth excels at driving solutions toward the origin, outperforming alternatives on CDF-corrected angular distance in 14 of 16 problem types (Figure 3). Balanced provides greater stability and tracks NSGA-II more closely on IGD metrics—likely because its symmetric fitness landscape encourages population diversity similar to crowding distance. Choose TrueNorth for aggressive minimization; choose Balanced when equitable trade-offs or interpretability are priorities.

Disconnected fronts. DTLZ7’s disconnected Pareto front poses a fundamental challenge for angular fitness: solutions on separate front segments may have similar angular distances despite occupying geometrically distinct regions. HFF’s single-pole selection pressure cannot simultaneously drive convergence toward multiple disconnected targets.

Scalability. We tested up to 1000 objectives; the $O(MN)$ complexity suggests scalability to even higher dimensions is computationally feasible. The practical limit is likely problem-dependent.

Normalization. Our implementation uses two-stage normalization: column-wise min-max scaling (making objectives comparable across different scales) followed by row-wise hyperspherical projection. This handles heterogeneous objective scales automatically. Results may be sensitive to normalization choice in domains with known objective bounds.

CDF correction assumptions. The CDF correction (Equation 6) assumes a uniform random baseline on the hypersphere. This is appropriate for measuring selection pressure relative to random search, but does not capture algorithm-specific biases in solution distributions.

8 Conclusion

We introduced hyperspherical fitness functions, a family of angular scalarization methods for many-objective optimization. By projecting objective vectors onto the unit hypersphere and measuring fitness as angular distance from a target direction, HFF provides a geometrically principled mechanism for ranking solutions—including those that Pareto dominance cannot distinguish.

Our experimental evaluation on DTLZ1–7 and WFG1–9 benchmarks from 10 to 1000 objectives reveals:

- HFF-TrueNorth delivers solutions orders of magnitude closer to the minimization target than NSGA-II on CDF-corrected metrics, winning 14 of 16 problem types.
- NSGA-II retains superiority on Pareto front coverage (IGD), confirming that HFF complements rather than replaces dominance-based methods.
- HFF-Balanced tracks NSGA-II on IGD more closely than expected, while providing equitable treatment of objectives.
- Selection complexity reduces from $O(MN^2)$ to $O(MN)$.

The concept of Arctic Circles—angular level sets on the hypersphere—provides both a selection mechanism and a fairness criterion: solutions on the same Arctic Circle represent equitable trade-offs where no objective is privileged. This geometric perspective may prove useful in multi-stakeholder optimization where fairness matters as much as performance.

Future directions. Several avenues warrant further investigation. First, HFF-TrueNorth and HFF-Balanced could serve as crowding distance operators within NSGA-II, combining Pareto-based selection with angular diversity maintenance. Second, real-world applications involving control loops and digital twins offer promising testbeds—domains where rapid scalarization and interpretable trade-offs may prove advantageous. Third, scaling and improving the hyperspherical approach itself remains open: better CDF approximations, GPU acceleration for billion-point evaluations, and theoretical analysis of convergence properties under HFF selection. Implementation available from the authors upon request.

Acknowledgments

We thank the developers of pymoo [4] for providing the multi-objective optimization framework used in our experiments. Claude Code was used to assist with code implementation and manuscript preparation. All conceptual contributions, experimental design, and scientific claims are the sole responsibility of the authors. Implementation available upon request.

A Implementation Details

The core hyperspherical fitness calculation in pseudocode. This reflects the actual implementation used to generate all experimental results:

HFF-Balanced.

Require: Objective vector $\mathbf{x} \in \mathbb{R}^m$

Ensure: Angular fitness $\theta \in [0, \pi]$

```

1:  $E \leftarrow \|\mathbf{x}\|^2$  {Total energy}
2: if  $E < \epsilon$  then
3:   return 0 {Perfect minimization}
4: end if
5:  $\mathbf{y} \leftarrow \mathbf{x} / \sqrt{E}$  {Unit projection}
6:  $\cos \theta \leftarrow \frac{1}{\sqrt{m}} \sum_{i=1}^m y_i$  {Dot with balanced pole}
7:  $\theta \leftarrow \arccos(\text{clamp}(\cos \theta, -1, 1))$ 
8: return  $\theta$ 
```

HFF-TrueNorth.

Require: Objective vector $\mathbf{x} \in \mathbb{R}^m$

Ensure: Angular fitness $\theta \in [0, \pi]$

```

1:  $E \leftarrow \|\mathbf{x}\|^2$  {Total energy}
2: if  $E < \epsilon$  then
3:   return 0 {Perfect minimization}
4: end if
5:  $\mathbf{y} \leftarrow \mathbf{x} / \sqrt{E}$  {Unit projection}
6:  $\bar{e} \leftarrow E/m$  {Normalized energy}
7:  $e \leftarrow \max(0, 1 - \bar{e})$  {Energy score  $\in [0, 1]$ }
8:  $s \leftarrow \sqrt{1 - e^2}$  {Scale factor}
9:  $\mathbf{y}' \leftarrow (s \cdot y_1, \dots, s \cdot y_m, e)$  {Augmented vector,  $\|\mathbf{y}'\| = 1$ }
10:  $\mathbf{p} \leftarrow (0, \dots, 0, 1)$  {North pole in  $\mathbb{R}^{m+1}$ }
11:  $\theta \leftarrow \arccos(\mathbf{y}' \cdot \mathbf{p}) = \arccos(e)$ 
12: return  $\theta$ 

```

Sign preservation. The projection preserves the sign of each objective coordinate. This generalization supports problems where objectives may take negative values. In our experiments on DTLZ and WFG benchmarks, all objective values are non-negative, so sign preservation has no effect on reported results.

B WFG Pareto Front Sampling

Standard implementations such as pymoo cannot generate WFG reference fronts beyond approximately 15 objectives because they enumerate $2^{2(m-1)}$ extreme points. We bypass this limitation through analytical sampling based on the Pareto front geometries defined by Huband et al. [15].

Algorithm 1 WFG1–2: Convex Scaled Simplex Sampling

Require: Objectives m , samples n
Ensure: Reference front $\mathbf{R} \in \mathbb{R}^{n \times m}$

```

1:  $\mathbf{W} \leftarrow n$  samples from Dirichlet( $\mathbf{1}_m$ )
2:  $R_{ij} \leftarrow 2j \cdot W_{ij}$  for all  $i, j$ 
3: return  $\mathbf{R}$ 

```

Algorithm 2 WFG3: Degenerate Linear Front Sampling

Require: Objectives m , samples n
Ensure: Reference front $\mathbf{R} \in \mathbb{R}^{n \times m}$

```

1:  $\mathbf{t} \leftarrow n$  samples from Uniform(0, 1)
2:  $\mathbf{R} \leftarrow \mathbf{0}^{n \times m}$ 
3:  $R_{i,1} \leftarrow 2(1 - t_i), R_{i,m} \leftarrow 2m \cdot t_i$  for all  $i$ 
4: return  $\mathbf{R}$ 

```

Algorithm 3 WFG4–9: Unit Hypersphere Sampling (Muller-Marsaglia [19])

Require: Objectives m , samples n
Ensure: Reference front $\mathbf{R} \in \mathbb{R}^{n \times m}$

```

1:  $\mathbf{Z} \leftarrow n \times m$  samples from  $\mathcal{N}(0, 1)$ 
2:  $Z_{ij} \leftarrow |Z_{ij}|$  for all  $i, j$  {Positive orthant}
3:  $R_{ij} \leftarrow Z_{ij} / \|\mathbf{Z}_i\|_2$  for all  $i, j$  {Normalize}
4: return  $\mathbf{R}$ 

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All experiments use $n = 10,000$ reference points per problem with a fixed random seed for reproducibility.

References

- [1] Johannes Bader and Eckart Zitzler. 2011. HypE: An Algorithm for Fast Hypervolume-Based Many-Objective Optimization. In *Evolutionary Computation*, Vol. 19. 45–76.
- [2] Nicola Beume, Carlos M. Fonseca, Manuel Lopez-Ibanez, Luís Paquete, and Jan Vahrenhold. 2009. On the complexity of computing the hypervolume indicator. *IEEE Trans. Evol. Comput.* 13, 5 (2009), 1075–1082.
- [3] Nicola Beume, Boris Naujoks, and Michael Emmerich. 2007. SMS-EMOA: Multiobjective Selection Based on Dominated Hypervolume. *European Journal of Operational Research* 181, 3 (2007), 1653–1669.
- [4] Julian Blank and Kalyanmoy Deb. 2020. pymoo: Multi-Objective Optimization in Python. *IEEE Access* 8 (2020), 89497–89509. doi:10.1109/ACCESS.2020.2990567
- [5] Dimo Brockhoff, Tobias Wagner, and Heike Trautmann. 2012. On the properties of the R2 indicator. In *GECCO 2012*. ACM, 465–472.
- [6] Ran Cheng, Yaochu Jin, Markus Olhofer, and Bernhard Sendhoff. 2016. A Reference Vector Guided Evolutionary Algorithm for Many-Objective Optimization. *IEEE Transactions on Evolutionary Computation* 20, 5 (2016), 773–791.
- [7] Carlos A. Coello Coello, Silvia González Brambila, Josué Figueroa Gamboa, Ma Guadalupe Castillo Tapia, and Raquel Hernández Gómez. 2020. Evolutionary multiobjective optimization: open research areas and some challenges lying ahead. *Complex & Intelligent Systems* 6 (2020), 221–236.
- [8] Carlos A. Coello Coello, Gary B. Lamont, and David A. Van Veldhuizen. 2007. *Evolutionary Algorithms for Solving Multi-Objective Problems* (2nd ed.). Springer.
- [9] Indraneel Das and John E. Dennis. 1997. A closer look at drawbacks of minimizing weighted sums of objectives for Pareto set generation. *Structural Optimization* 14, 1 (1997), 63–69.
- [10] Kalyanmoy Deb and Himanshu Jain. 2014. An Evolutionary Many-Objective Optimization Algorithm Using Reference-Point-Based Nondominated Sorting Approach, Part I: Solving Problems With Box Constraints. *IEEE Transactions on Evolutionary Computation* 18, 4 (2014), 577–601.
- [11] Kalyanmoy Deb, Amrit Pratap, Sameer Agarwal, and T. Meyarivan. 2002. A Fast and Elitist Multiobjective Genetic Algorithm: NSGA-II. *IEEE Transactions on Evolutionary Computation* 6, 2 (2002), 182–197.
- [12] Kalyanmoy Deb, Lothar Thiele, Marco Laumanns, and Eckart Zitzler. 2005. Scalable Test Problems for Evolutionary Multiobjective Optimization. *Evolutionary Multiobjective Optimization* (2005), 105–145.
- [13] Marco Farina and Paolo Amato. 2004. A Fuzzy Definition of "Optimality" for Many-Criteria Optimization Problems. *IEEE Transactions on Systems, Man, and Cybernetics - Part A* 34, 3 (2004), 315–326.
- [14] Michael Grieves and John Vickers. 2017. Digital Twin: Mitigating Unpredictable, Undesirable Emergent Behavior. In *Transdisciplinary Perspectives on Complex Systems*.
- [15] Simon Huband, Philip Hingston, Luigi Barone, and Lyndon While. 2006. A Review of Multiobjective Test Problems and a Scalable Test Problem Toolkit. *IEEE Transactions on Evolutionary Computation* 10, 5 (2006), 477–506.
- [16] Mikkel T. Jensen. 2003. Reducing the Run-Time Complexity of Multiobjective EAs: The NSGA-II and Other Algorithms. *IEEE Transactions on Evolutionary Computation* 7, 5 (2003), 503–515.
- [17] Joshua Knowles and David Corne. 2007. Quantifying the effects of objective space dimension in evolutionary multiobjective optimization. In *EMO 2007 (LNCS, Vol. 4403)*. Springer, 757–771.
- [18] Bingdong Li, Jinlong Li, Ke Tang, and Xin Yao. 2015. Many-objective evolutionary algorithms: A survey. *Comput. Surveys* 48, 1 (2015).
- [19] Mervin E. Muller. 1959. A Note on a Method for Generating Points Uniformly on N-dimensional Spheres. *Commun. ACM* 2, 4 (1959), 19–20.
- [20] Frans P. B. Osinga. 2007. *Science, Strategy and War: The Strategic Theory of John Boyd*. Routledge.
- [21] Jorge Stolfi. 1991. *Oriented Projective Geometry: A Framework for Geometric Computations*. Academic Press, Boston.
- [22] Fei Tao and Qinglin Qi. 2019. Digital Twin Driven Smart Manufacturing. *IEEE Access* (2019).
- [23] Qingfu Zhang and Hui Li. 2007. MOEA/D: A Multiobjective Evolutionary Algorithm Based on Decomposition. *IEEE Transactions on Evolutionary Computation* 11, 6 (2007), 712–731.
- [24] Weijia Zheng, Yang Nan, and Hisao Ishibuchi. 2025. *Weak Pareto Boundaries and Their Negative Impact on Evolutionary Multi-Objective Optimization*. arXiv:2505.13854 [cs.NE]
- [25] Eckart Zitzler. 1999. *Evolutionary Algorithms for Multiobjective Optimization: Methods and Applications*. Ph.D. Dissertation. ETH Zurich.
- [26] Eckart Zitzler and Simon Künzli. 2004. Indicator-Based Selection in Multiobjective Search. In *Proc. Parallel Problem Solving from Nature (PPSN)*. Springer, 832–842.
- [27] Eckart Zitzler, Lothar Thiele, Marco Laumanns, Carlos M. Fonseca, and Viviane Grunert da Fonseca. 2003. Performance assessment of multiobjective optimizers: An analysis and review. *IEEE Trans. Evol. Comput.* 7, 2 (2003), 117–132.